

An H_{FD} Group Outside the Erschler–Ozawa Random-Walk Criterion

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Status and exact sources

Status: `literature_already_answered` for one question in the introduction of Erschler–Ozawa, arXiv:1609.08585 [1].

The source passage is on PDF page 3, immediately after Corollary 2.5. The answer is stated on PDF page 2 of the distinct later paper of Brioussell–Zheng, arXiv:1706.00707 [2], and its construction is detailed in Subsection 3.3.

Source question

For a symmetric finitely supported generating probability measure μ on a finitely generated group G , Corollary 2.5 of [1] proves that G has Shalom’s property H_{FD} if either

$$\liminf_{n \rightarrow \infty} \left\| \mu^{*n} - \mu^{*(1+\delta)n} \right\|_1 < 2 \quad \text{for some } \delta > 0,$$

or

$$\limsup_{n \rightarrow \infty} \mu^{*n}(B_G(c\sqrt{n})) > 0 \quad \text{for every } c > 0.$$

The authors then record the existence question: they know no group satisfying H_{FD} while failing the assumption of Corollary 2.5.

Exact later answer

Brioussell–Zheng first recall precisely this Erschler–Ozawa criterion and then state on PDF page 2:

Examples of groups with property H_{FD} not satisfying the assumption of Erschler–Ozawa’s corollary can be obtained from their construction.

Here this sentence is paraphrased only by suppressing cross-references; the mathematical assertion is exact.

The construction in Subsection 3.3 is a diagonal product Δ formed from finite wreath-product factors. Its FC-center $Z^{\text{FC}}(\Delta)$ has quotient

$$\Delta/Z^{\text{FC}}(\Delta) \cong (A \times B) \wr \mathbb{Z},$$

with $A \times B$ finite. The weakly mixing harmonic-cocycle factorization used there shows that Δ has the stronger property H_{T} , hence H_{FD} . Varying the scales in the same construction gives the random-walk behavior needed to lie outside the sufficient hypothesis. Thus the later paper supplies the existence examples requested in [1]; this is a literature answer, not a new theorem of the present run.

Scope limitation

This identification answers only the existence question following Corollary 2.5. It does not settle the source's separate comparison with Ozawa's shifted-convolution condition, either question about Følner balls, the geometric detectability of torsion, or the status of $(\mathbb{Z}/2\mathbb{Z}) \wr \mathbb{Z}^2$. Indeed [2] still lists the last problem as open.

References

- [1] A. Erschler and N. Ozawa, *Finite-Dimensional Representations constructed from Random Walks*, arXiv:1609.08585 (2016; revised 2017).
- [2] J. Brieussel and T. Zheng, *Shalom's property H_{FD} and extensions by $\mathbb{Z}$ of locally finite groups*, arXiv:1706.00707 (2017).